

Delegation Intelligence: A Task-Difficulty, Human-Intervention, and Reliability Framework for Autonomous AI Agents and Organizations

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Task Difficulty x Human Cognitive Intervention x Verified Reliability

Abstract

AI autonomy is often described with phrases such as assistant, agent, autonomous agent, or fully autonomous system. These labels mix together at least three different questions: how difficult a task is, how much human cognition must remain inside the execution loop, and how reliably the system succeeds. A model that independently completes trivial tasks is not equivalent to one that independently completes expert or frontier tasks; conversely, a system that solves frontier problems only while a human supplies the key plans and corrections is highly capable but weakly delegable. This paper proposes Delegation Intelligence (DL) as a distinct axis of intelligence: the empirically measured ability of an individual artificial intelligence or an intelligent organization to accept a delegated task, goal, mission, or bounded charter and produce a verified result under a specified human-intervention budget.

The framework separates four concepts that are frequently conflated: task difficulty, human cognitive intervention, governance intervention, and reliability. It introduces a task scale T0-T6, a human cognitive-intervention scale H0-H5, and a Delegation Intelligence scale DL0-DL6 plus DL Ω . DL0 systems execute atomic instructions under continuous direction; DL3 systems accept outcomes and autonomously determine task strategy; DL4 systems reliably complete complex expert tasks with exception-only or no cognitive intervention; DL5 systems solve problems whose solution is not known in advance; DL6 systems accept missions and generate the goals, projects, and tasks required to accomplish them; and DL Ω systems operate under a bounded charter and can identify new worthwhile problems and missions without receiving each one from a human.

A central contribution is the delegation frontier, $F_A(h,p)$, defined as the highest task difficulty that system A completes with success probability at least p under human cognitive-intervention budget h . This turns delegation from a rhetorical label into a longitudinal performance surface. The framework further distinguishes cognitive assistance from governance approval: a human safety gate can remain in the loop without implying that the human supplied the missing intelligence. The same formalism is extended from individuals (DL-I) to organizations (DL-O), where the delegated unit is a persistent coordination structure rather than a single decision lineage. Finally, the paper relates Delegation Intelligence to developmental intelligence levels, organizational intelligence, and substrate-indexed recursive self-improvement, arguing that these are complementary rather than interchangeable coordinates. The result is a falsifiable framework for answering a practical question: How difficult a goal can a person hand to this intelligence, how little must the person think for it, and how often does the result survive independent verification?

Keywords: delegation intelligence; AI agents; autonomy; human-AI interaction; task difficulty; human intervention; agent evaluation; organizational intelligence; autonomous systems; recursive self-improvement

1. Introduction

The practical value of an intelligent system is not exhausted by what it can answer in a benchmark or what it can do when an expert continuously steers it. In work, science, engineering, and administration, a central question is delegation: what can a person hand over and expect to come back correctly completed?

Existing automation research has long recognized that autonomy is not binary. Sheridan and Verplank described graded allocations of decision and action between human and computer. Parasuraman, Sheridan, and Wickens later decomposed automation into information acquisition,

information analysis, decision and action selection, and action implementation, each of which can be automated to different degrees. Endsley and Kaber showed that automation level changes not only system performance but also operator workload and situation awareness. These traditions provide a strong foundation for understanding how much of an operation is assigned to automation. Modern AI agents create a different measurement problem: the delegated object may be an entire goal or project whose internal task structure is initially unspecified.

For such agents, autonomy alone is insufficient. Consider two systems. System A independently renames files, formats reports, and executes fixed scripts with no human help. System B can implement a multi-module software feature, diagnose failures, revise its plan, verify the result, and ask for help only when a genuinely unavailable decision is encountered. Both can be described as "autonomous," but the delegated cognitive burden is radically different. Likewise, a third system may help solve a difficult scientific problem but only because a human researcher chooses the hypotheses, diagnoses each failed experiment, and supplies the next step. Its task capability may be high while its delegation intelligence remains limited.

This paper therefore treats delegation as a joint function of difficulty, intervention, and reliability. A DL claim is not "the agent can act by itself." It is a statement of the form:

On a predeclared family of tasks of difficulty T , the system reaches independently verified success with probability at least p while requiring no more than intervention budget h .

This formulation makes three changes to the usual discourse. First, it prevents trivial-task autonomy from being mistaken for broad autonomy. Second, it prevents human-steered problem solving from being mistaken for independent delegation. Third, it makes failure frequency part of the level rather than an afterthought.

The framework is intended as a complement to, not a replacement for, other intelligence scales. Developmental intelligence asks what the system can persistently learn or change about itself. Organizational intelligence asks what a coordination structure can persistently learn or change about itself. Substrate-indexed recursive self-improvement asks which software or physical substrate an improvement loop can autonomously modify. Delegation Intelligence asks a different question: how much difficult work can a human remove from their own cognitive loop and place into the system while preserving verified outcomes?

2. Related Work and the Missing Coordinate

2.1 Levels of automation

Sheridan and Verplank's supervisory-control framework established a continuum between fully manual action and increasingly autonomous computer decision and execution. Parasuraman, Sheridan, and Wickens generalized this tradition by separating automation across four functional stages: acquisition, analysis, decision/action selection, and implementation. Their central insight remains important for agentic AI: automation is multidimensional, and high automation at one stage does not imply high automation at every stage.

Delegation Intelligence retains this insight but changes the unit of analysis. Classical automation often begins with a known function and asks how much of that function should be automated. DL begins with a delegated task or goal and asks how much of the cognitive work needed to transform that goal into a verified outcome is supplied by the artificial system.

2.2 Human performance under automation

Human-factors work also warns that autonomy changes the human's role rather than simply deleting it. Endsley and Kaber demonstrated tradeoffs among performance, workload, and situation awareness across levels of automation. Klein and colleagues emphasized that automation acting as a team member must remain predictable, directable, and coordinated with others. These results motivate two DL requirements. First, the framework should report the kind of intervention still required, not merely its frequency. Second, organizational delegation should include coordination quality rather than treating several parallel agents as automatically equivalent to one coherent organization.

2.3 Capability, generality, and autonomy

Recent AI level frameworks distinguish capability from autonomy. This is necessary but still incomplete for practical delegation. Autonomy is usually treated as a deployment property: how independently a system can operate. Delegation requires an additional difficulty-conditioned measurement: independent at what level of task, and with what probability of verified success? The same agent may be highly autonomous on T1 tasks and weakly autonomous on T4 tasks.

2.4 The missing coordinate

The proposed DL framework therefore adds a coordinate not captured by task performance alone or by a generic autonomy label. It measures the boundary between work that can be handed over and work that still requires human cognition.

3. Formal Object of Delegation

3.1 Delegated task

A delegated task is represented as

$$\tau = \langle G, X, C, R, V, H \rangle$$

where G is the goal, X the initial state and available evidence, C the constraints, R the available resources and tools, V the acceptance/verification criterion, and H the time horizon or stopping condition. A task family T is a distribution over such tasks rather than a single example.

The human may specify more or less of this tuple. At low DL levels the human may provide the full procedure. At higher DL levels the human increasingly supplies only G, C, and V; the system must infer or construct subgoals, plans, tool sequences, evidence needs, and recovery strategies.

3.2 Delegation episode

A delegation episode starts when responsibility for a task is transferred to the system and ends when the system either produces a result, refuses/escalates, or fails. During the episode, every human intervention is recorded with a type, duration, and effect.

3.3 Verified success

A result counts as success only if it satisfies a criterion declared independently of the system's self-report. Depending on the domain, this can be a deterministic test, held-out dataset, externally inspected artifact, replicated experiment, or other evidence-bearing evaluation. The agent saying that it finished does not establish successful delegation.

4. Task Difficulty: T0-T6

Task difficulty must be defined independently of the particular system being evaluated. Otherwise a task becomes "easy" merely because one agent solved it.

Level	Name	Operational character
T0	Atomic	One explicit operation; negligible planning or uncertainty.
T1	Routine	Familiar short task with a known procedure and clear acceptance condition.
T2	Multi-step	Several dependent operations; modest planning; known goal and success criterion.
T3	Complex	Nontrivial decomposition, tool choice, state tracking, error recovery, and replanning.
T4	Expert	Long-horizon or ambiguous task requiring substantial domain reasoning, tradeoffs, and verification design.
T5	Frontier	The solution method or relevant knowledge is not known in advance; discovery or invention is required.
T6	Open mission	The system must maintain a mission across changing conditions and generate substantial portions of the project/task structure itself.

4.1 Difficulty should be a vector before it is a band

The T0-T6 labels are reporting bands. Internally, a task should be profiled along dimensions such as decomposition depth, horizon, uncertainty, tool diversity, environmental change, domain expertise, novelty, verification difficulty, coordination burden, and cost of error. Two tasks at the same T band can be difficult for different reasons.

Let

$$d(\tau) = \langle d_{\text{dec}}, d_{\text{hor}}, d_{\text{unc}}, d_{\text{tool}}, d_{\text{nov}}, d_{\text{ver}}, d_{\text{coord}}, d_{\text{risk}} \rangle.$$

A benchmark may map this vector to a T band using predeclared thresholds or calibrated human/expert judgments. The vector should remain available so that a system cannot hide a narrow weakness behind an average difficulty score.

4.2 Difficulty is not duration

A task can take a long time and still be routine. Conversely, a short task can be intellectually difficult. Duration may correlate with difficulty but is not constitutive of it.

5. Human Cognitive Intervention: H0-H5

Delegation depends on how much human cognition remains necessary during execution.

Level	Name	Meaning
H0	None	No human cognitive contribution is required after delegation.
H1	Exception-only	Human cognition is needed only for rare, explicitly exceptional states.
H2	Occasional	Human decisions or corrections are needed at important checkpoints.

Level	Name	Meaning
H3	Periodic	The system can work for meaningful stretches but regularly needs human direction.
H4	Frequent	Human planning/correction repeatedly determines the next steps.
H5	Continuous	The human effectively supplies the task strategy step by step.

5.1 Cognitive intervention versus governance intervention

The framework separates cognitive intervention from governance intervention.

A cognitive intervention supplies information or reasoning the system needed in order to solve the task: diagnosing its mistake, choosing the correct strategy, resolving an ambiguity it could have resolved from available evidence, supplying a missing intermediate result, or rescuing a failed plan.

A governance intervention authorizes or blocks an action for reasons external to the system's cognitive competence: approving expenditure, allowing publication, signing a deployment, granting access, or halting activity under a safety policy.

Therefore:

human can intervene != human must think for the system.

A system can remain H0 or H1 in cognitive intervention while still operating under strong human governance. Both should be logged, but only the former determines DL.

5.2 Intervention types

Each intervention should be labeled at minimum as approval, information supply, planning assistance, correction, rescue, or environment/permission resolution. The label matters because one deep rescue can represent more missing intelligence than many routine approvals.

6. Reliability and the Delegation Frontier

Autonomy without reliability is not useful delegation. A system that independently attempts T5 tasks but succeeds on 2% of them is not DL5.

Define

$$S_A(T,h) = P(\text{success} \mid \text{system A, task family T, intervention budget h}).$$

For a required success threshold p , define the delegation frontier:

$$F_A(h,p) = \max \{ T : S_A(T,h) \geq p \}.$$

$F_A(h,p)$ is the hardest task band the system reliably completes under the allowed human cognitive-intervention budget.

A stronger representation is a surface rather than a single value: different h and p values produce different frontiers. A research agent may have $F_A(H2,0.90)=T4$ but only $F_A(H0,0.90)=T2$. This is much more informative than calling the system "autonomous."

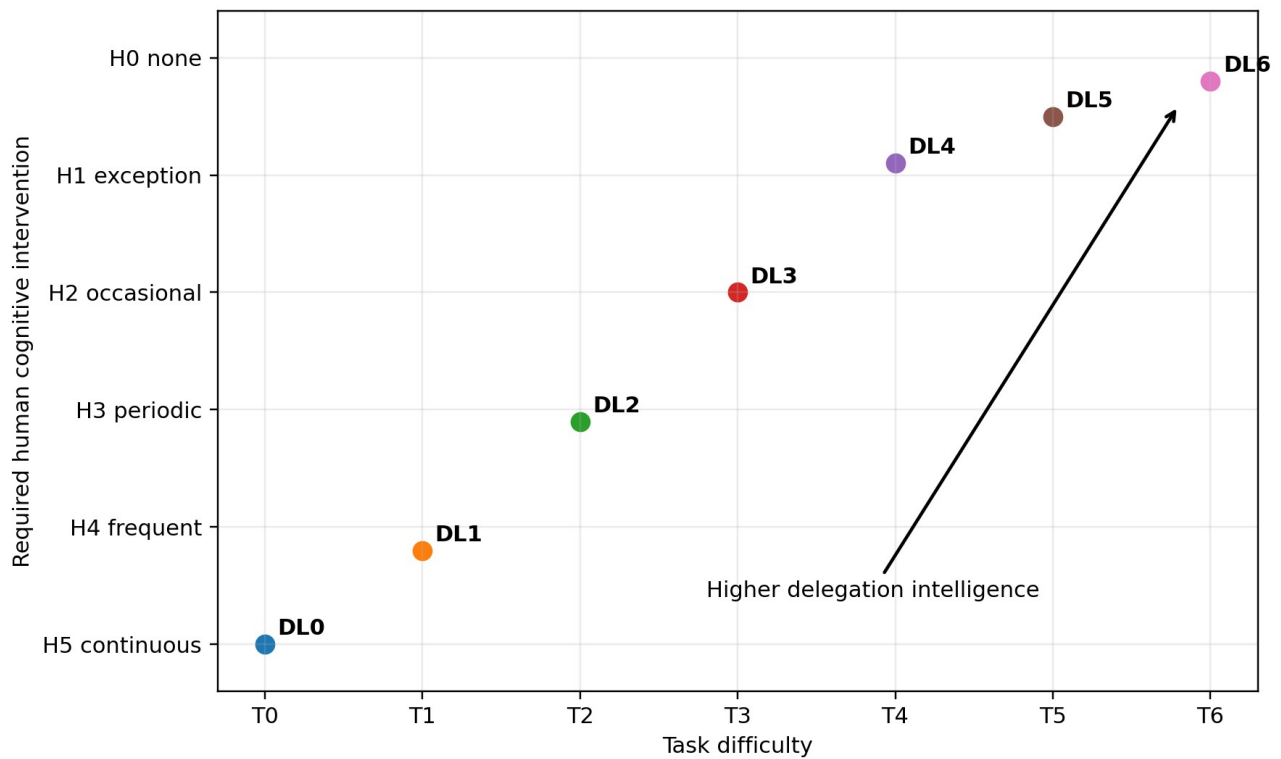


Figure 1. Delegation Intelligence occupies a joint space of task difficulty and required human cognitive intervention. Higher levels require both harder tasks and less cognitive assistance.

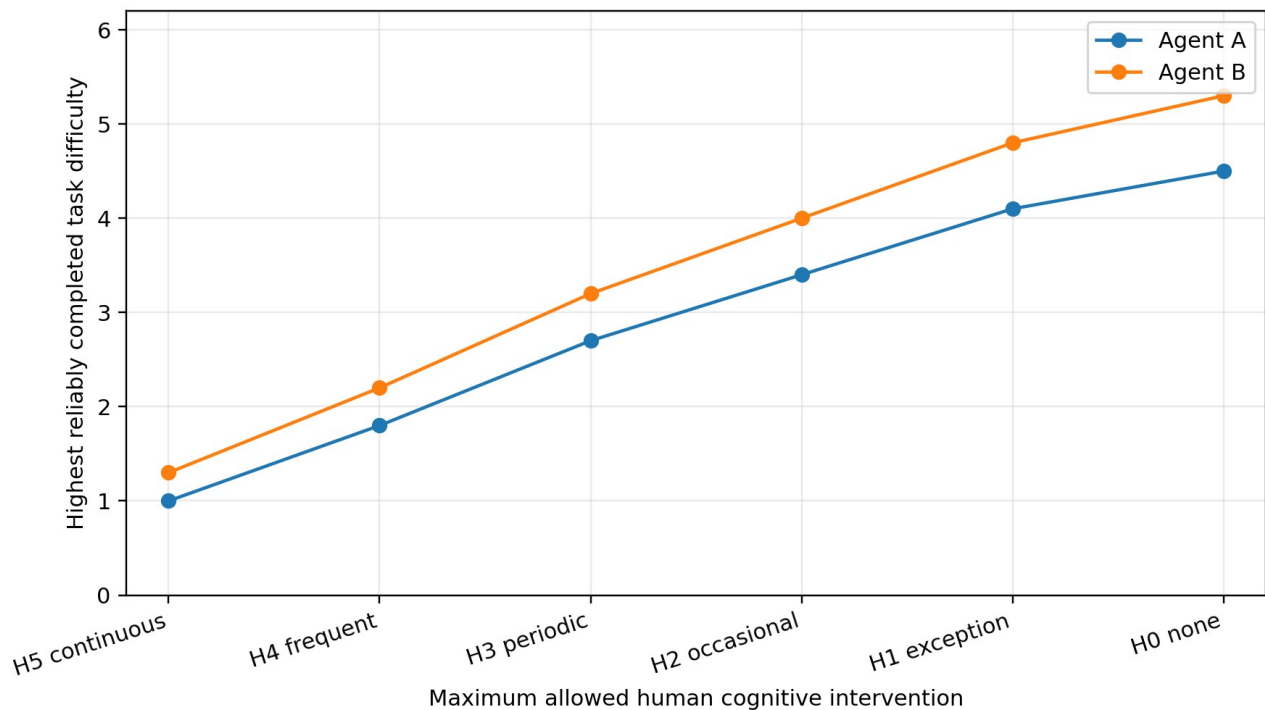


Figure 2. A delegation frontier compares systems by the highest task difficulty they complete reliably under different intervention budgets. The frontier is a profile, not a single benchmark score.

7. Individual Delegation Intelligence: DL0-DL Ω

7.1 DL0 - Instruction-Following Intelligence

Human contribution: specifies nearly every operation.

Agent contribution: executes an atomic instruction correctly.

Typical region: T0 with H5-H4.

DL0 is useful and should not be treated pejoratively. Deterministic execution, extraction, transformation, and tool invocation can be highly valuable. What DL0 lacks is independent task control: the human remains the planner.

Transition test DL0 -> DL1: the system must accept a complete small task rather than a single operation and correctly choose at least a short sequence of actions.

7.2 DL1 - Assisted Task Intelligence

Human contribution: supplies a small, well-specified task and frequent supervision.

Agent contribution: constructs and executes a short local plan.

Typical region: T1 with H4-H3.

The system can choose routine steps and tools but frequently depends on correction. The human still carries much of the task model.

Transition test DL1 -> DL2: on a predeclared family of T2 tasks, the agent must sustain a multi-step plan, preserve state, recover from ordinary errors, and meet the success threshold with no more than periodic intervention.

7.3 DL2 - Supervised Delegation Intelligence

Human contribution: specifies the task and checks progress periodically.

Agent contribution: decomposes a routine multi-step task, chooses tools, tracks state, and produces a verifiable result.

Typical region: T2 with H3-H2.

DL2 is the first level at which delegation becomes economically meaningful over nontrivial tasks. The human is no longer specifying every next action.

7.4 DL3 - Independent Task Intelligence

Human contribution: specifies the desired outcome, constraints, and acceptance criteria.

Agent contribution: determines the task strategy.

Typical region: T3 with H2-H1.

A DL3 system can create its own task graph, select and sequence tools, detect many failures, replan, and integrate evidence. Human help is occasional rather than structurally required.

The defining change is the transfer of procedural responsibility. At DL2 the human still delegates a recognizable multi-step task. At DL3 the human can state the outcome and let the system determine how to achieve it.

7.5 DL4 - Autonomous Complex-Task Intelligence

Human contribution: supplies a complex objective and constraints.

Agent contribution: performs long-horizon planning, execution, adaptation, verification, and recovery with exception-only or no cognitive intervention.

Typical region: T4 with H1-H0.

DL4 should be a demanding threshold. The task family must contain ambiguity, unexpected failures, and verification work. Merely running a long predetermined workflow does not qualify.

7.6 DL5 - Autonomous Expert / Frontier Intelligence

Human contribution: supplies a hard problem whose solution is not already known.

Agent contribution: discovers or invents the missing solution method or knowledge and validates it.

Typical region: T5 with H1-H0.

At DL4, the solution path is difficult. At DL5, the relevant path or knowledge is initially unknown. This requires structured uncertainty, hypothesis formation or method invention, discriminating evidence, and external validation.

A plausible-sounding novel answer is not DL5. The system must produce knowledge or a solution that survives an evaluation independent of its own confidence.

7.7 DL6 - Mission-Level Autonomous Intelligence

Human contribution: supplies a mission rather than a task list.

Agent contribution: derives goals, projects, tasks, priorities, and execution strategies over a long horizon.

Typical region: T6 with H1-H0.

The key transition is from task autonomy to agenda construction inside a mission. A DL6 system may create and retire subprojects, allocate resources, detect that the original plan is obsolete, and generate new work required by the mission.

DL6 does not imply unrestricted authority. A system may autonomously determine what work is cognitively necessary while still requiring external authorization for consequential actions.

7.8 DL Ω - Open-Ended Delegation Intelligence

Human contribution: supplies a bounded charter, domain, or enduring purpose rather than each mission.

Agent contribution: observes the domain, identifies worthwhile new problems, formulates missions, creates tasks, solves them, learns, and repeats.

DL Ω is deliberately separated from DL6 because the system now generates not only tasks within a mission but missions within a charter. The cycle is:

Charter -> Observe -> Identify opportunity/problem -> Select mission -> Create tasks -> Execute -> Verify -> Learn -> Identify deeper opportunities.

DL Ω should be treated as provisional evidence rather than something established by one long run. Open-endedness cannot be proved by a finite benchmark. Strong evidence would require repeated mission generation across changing conditions, independent validation, sustained usefulness, low cognitive-intervention burden, and explicit protection against self-redefinition of success.

8. Organizational Delegation Intelligence

The same delegation question applies to organizations, but the delegated unit is different. An individual agent has one persistent decision lineage. An intelligent organization contains multiple persistent members plus routing, roles, memory, verification, and authority structures.

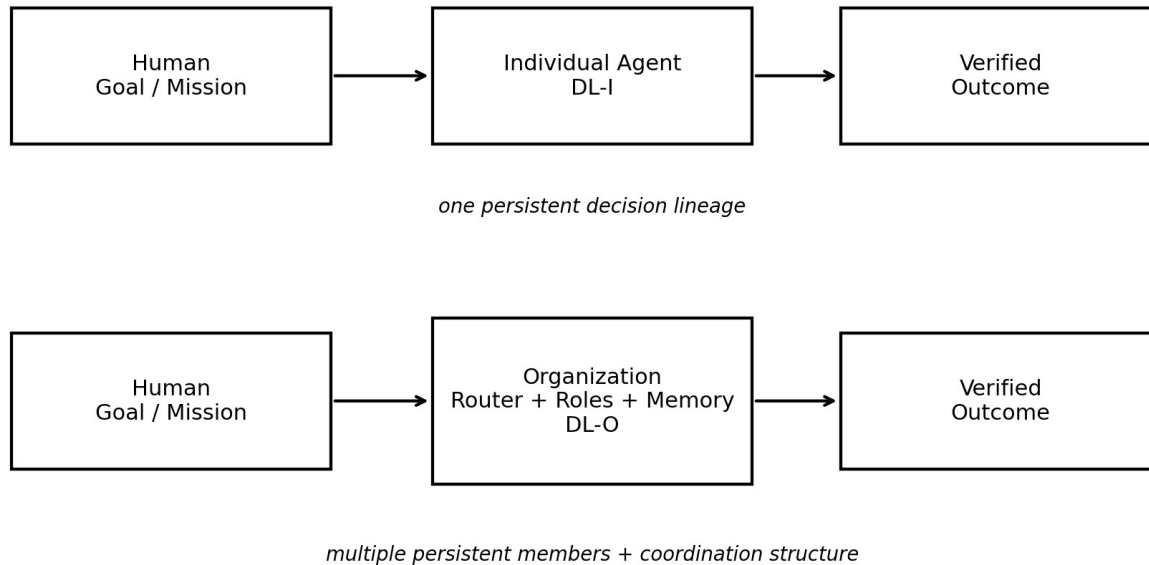


Figure 3. DL-I measures delegation to one persistent individual intelligence. DL-O measures delegation to a coordination structure whose result depends on multiple persistent members and their organization.

8.1 O-DL0 - Human-routed organization

Humans assign the work to each member and coordinate the steps. The organization is a collection of tools or specialists, not an independently delegable unit.

8.2 O-DL1 - Assisted organizational delegation

The organization routes simple work and aggregates results, but humans frequently resolve task decomposition and coordination.

8.3 O-DL2 - Supervised project organization

A human delegates a routine project. The organization decomposes work, assigns specialists, preserves shared state, and produces a result with periodic human management.

8.4 O-DL3 - Independent project organization

A human states the project outcome. The organization creates the project structure, assigns members, manages dependencies, reconciles evidence, and reaches verification with only occasional human cognitive intervention.

8.5 O-DL4 - Autonomous complex-program organization

A human delegates a complex program. The organization manages multiple parallel workstreams, reallocates resources, changes coordination patterns, invokes independent verification, and resolves most internal failures without human cognitive rescue.

8.6 O-DL5 - Autonomous discovery organization

A human provides an unsolved problem. The organization constructs its own research or engineering program and the discovery depends on the division of cognitive labor: no single member need contain the whole solution path.

8.7 O-DL6 - Mission organization

A human supplies a mission. The organization generates projects and agendas, creates or selects specialist roles, manages long-horizon state, verifies work, and closes or replaces projects as mission conditions change.

8.8 O-DL Ω - Charter-level organization

The organization operates under a bounded charter and repeatedly identifies worthwhile missions, constructs the organizations or roles needed for them, produces independently validated outcomes, and learns which organizational forms work best. This is open-ended organizational delegation rather than merely a large set of autonomous agents.

9. Measuring Human Intervention

9.1 Intervention Load

A first metric is Human Cognitive Intervention Load (HCIL):

$$\text{HCIL} = \text{human cognitive effort required} / \text{total cognitive effort required by the completed task.}$$

HCIL is difficult to measure directly, so implementations should report observable proxies: intervention count, human cognitive minutes, number of human-generated next-step decisions, number of human corrections, number and depth of rescues, and the fraction of task branches whose strategy was selected by the human.

9.2 Critical Intervention Depth

Interventions should also be scored by criticality. An approval that says "proceed" is shallow. A rescue that supplies the central missing idea is deep. Define Critical Intervention Depth (CID) as the highest-level task decision supplied by a human during the episode:

- CID0: administrative/governance only;
- CID1: local factual clarification;
- CID2: local action correction;
- CID3: subtask strategy;
- CID4: overall task strategy;
- CID5: missing expert insight or discovery;
- CID6: mission/agenda definition beyond the delegated charter.

A system should not achieve a DL level if a human intervention at or above the defining cognitive layer was necessary for success.

9.3 Intervention timing

The same intervention has different meaning if it is proactive versus rescue. Human input supplied before the system encounters the relevant uncertainty may constitute hidden scaffolding. Benchmarks should therefore record when the information was supplied and whether it was available in the original task specification.

10. Anti-Inflation Evidence Rules

Delegation claims are vulnerable to inflation. The following rules are intended to make levels falsifiable.

Rule 1 - Independence from triviality. No high DL claim can be established on low-difficulty tasks, regardless of how autonomous execution is.

Rule 2 - Independence from one heroic run. Levels are assigned over task families with confidence intervals, not demonstrations.

Rule 3 - Human work must be counted. Planning, correction, hidden prompt engineering, expert hints, and rescue are interventions even if delivered through a chat interface rather than a formal control panel.

Rule 4 - Governance is reported separately. Required approval does not automatically lower DL if the approval contributes no missing cognition.

Rule 5 - Self-reported completion is insufficient. Success requires external or machine-grounded verification.

Rule 6 - Tool availability is part of the test condition. A system can use tools, but the tool set and external services must be recorded so results can be compared.

Rule 7 - Resource normalization matters. Very large increases in compute, latency, or monetary cost can improve success while reducing practical delegation value. Report them separately.

Rule 8 - Refusals and escalations are outcomes. A correct escalation can be preferable to a confident failure, but frequent escalation lowers the frontier at H0/H1.

Rule 9 - Task difficulty is externally calibrated. The evaluated system does not define the difficulty of the tasks used to level it.

Rule 10 - Mission-level claims require longitudinal evidence. DL6 and DL Ω require persistence through changing state, not a single long prompt.

11. Benchmark Design

A serious DL benchmark should be longitudinal and intervention-aware.

11.1 Task families

For each T band, construct multiple task families with distinct difficulty vectors. T3, for example, should include software, research, data, planning, and tool-use tasks whose complexity arises from different combinations of horizon, uncertainty, and verification burden.

11.2 Intervention protocol

Run each system under several intervention budgets: H0, H1, H2, and when useful H3. Human evaluators must follow a written policy specifying what assistance is permitted at each budget.

11.3 Success measurement

Each task needs a predeclared external criterion. A benchmark should not ask the same model that performed the work to decide whether it succeeded.

11.4 Censoring and escalation

If the agent correctly identifies that a task cannot be completed under its available evidence or permissions, the result should be recorded as escalation rather than failure-by-hallucination. However, escalation still means the task was not autonomously completed, so it affects the frontier.

11.5 Reporting form

A minimum report should include:

DL profile = {F_A(H0,p), F_A(H1,p), F_A(H2,p), HCIL, CID distribution, success CI, cost, latency}.

The scalar label DL_n should be a compact summary of the underlying profile.

12. Transition Tests

Transition	Minimum evidence
DL0 -> DL1	Complete T1 task from goal-level request, not step-by-step instruction.
DL1 -> DL2	Complete T2 task family with persistent multi-step state and no more than H3-H2 intervention.
DL2 -> DL3	Given outcome/constraints, autonomously construct and execute task strategy on T3 with at most H2-H1 intervention.
DL3 -> DL4	Sustain T4 work through ambiguity, unexpected failures, replanning, and verification at H1-H0 with required reliability.
DL4 -> DL5	Solve T5 problems whose solution method is initially unknown and obtain independent validation at H1-H0.
DL5 -> DL6	Accept mission, autonomously generate and manage multiple projects/tasks over changing conditions, and meet mission criteria at H1-H0.
DL6 -> DL Ω	Under a bounded charter, repeatedly identify valuable new missions, complete them, and demonstrate independently validated utility without routine human mission generation.

13. Relationship to Other Intelligence Axes

Delegation Intelligence should remain independent from developmental and organizational evolution levels.

An agent can be I1 but DL4: it has a powerful fixed cognitive architecture and persistent memory, does not learn or self-improve, yet autonomously completes difficult tasks.

Another can be I3 but DL2: it can improve parts of its own cognition under evaluation but remains unreliable at long-horizon delegated tasks.

Similarly, an organization can be organizationally adaptive while still requiring frequent human management, or be structurally simple but composed of strong agents that make it highly delegable on a narrow mission.

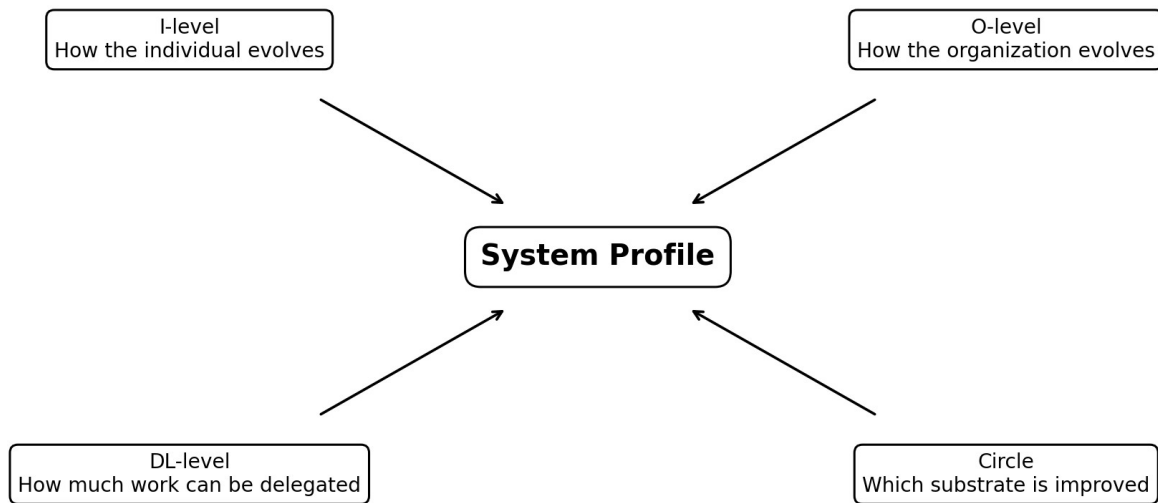


Figure 4. Delegation Intelligence is complementary to developmental I-levels, organizational O-levels, and substrate-circle completion. None substitutes for the others.

A compact system profile can therefore report:

$$A = (I_i, O_j, DL-I_k, DL-O_m, \lambda_vector, authority, resources).$$

The purpose is not to maximize the number of coordinates but to prevent one word - "intelligence" - from hiding substantively different properties.

14. Integration with Persistent LLM Agent Architectures

A persistent LLM agent can instrument DL directly. A long-lived worker should record every delegated task, its predeclared difficulty band, the intervention policy, plan/replan history, tool use, verification outcome, human interventions, and final result. This is compatible with architectures that separate the persistent brain from transient executors and independent verification.

A minimal episode record can include:

```

{
  "task_id": "tsk_...",
  "task_band": "T3",
  "difficulty_vector": {"horizon": 3, "uncertainty": 2, "verification": 3},
  "intervention_budget": "H1",
  "human_interventions": [
    {"type": "approval", "cognitive": false, "minutes": 0.2}
  ],
  "agent_replans": 4,
  "verification": "pass",
  "result": "success",
  "latency_seconds": 4200,

```

```
"compute_cost": 12.4
}
```

For a SARSI-style worker, this measurement naturally attaches to episodic memory, durable plans, exact context replay, pre-action forecasts, independent verification, and post-run consolidation. The existing implementation principle that the persistent worker plans, a transient executor acts, and an independent mechanism verifies is especially useful because it makes the boundary between delegation and self-certification explicit. The DL machinery should not widen authority; it measures cognitive delegation under whatever authority policy exists.

15. Individual and Organizational DL in Practice

15.1 Why earlier levels should remain in high-level systems

Higher DL does not make lower-DL components obsolete. Mature systems should decompose work. Atomic reliable procedures belong in DL0/DL1 executors; complex planning belongs in higher-DL agents; independent verification may deliberately use simpler, constrained components. This reduces cost and limits error propagation.

15.2 Delegation compression

A valuable property of intelligence is converting once-difficult work into routines. A T4 problem solved repeatedly may become a T2 procedure after the system creates tools, templates, checks, and knowledge. Delegation therefore changes both the agent and the task environment.

To preserve comparability, task difficulty should be calibrated against the current externally available environment. When the system itself creates a reusable tool that makes the family easier, report both the original difficulty and the post-tool difficulty. The tool is evidence of intelligence; silently relabeling the task as always easy would erase the achievement.

15.3 Organizational advantage

Organizations can increase delegation capacity through specialization, parallelism, independent verification, and persistent shared state. O-DL should therefore be evaluated on tasks whose successful completion genuinely depends on coordination, not merely on choosing the best single member.

16. Failure Modes

16.1 Autonomy theater

A system can appear autonomous because humans precompute detailed prompts, workflows, or hidden scaffolds. If those artifacts carry the missing planning burden, they must be counted as part of the human contribution or fixed test environment.

16.2 Approval inflation

Conversely, a system can appear less delegable merely because policy requires approvals. This is why governance and cognition must be separated.

16.3 Easy-task saturation

Success rates near 100% on T0/T1 tasks reveal little about T3/T4 delegation. Benchmarks must move along the difficulty axis.

16.4 Silent rescue

Humans often intervene conversationally in ways that are not labeled as help: "maybe check X," "that doesn't look right," or "use method Y." These can be substantial cognitive contributions and should be logged.

16.5 Verification leakage

If the system sees the exact held-out evaluator and can optimize directly for it, high success may represent benchmark gaming rather than robust delegation. Evaluation ownership and task-generation provenance should be protected.

16.6 Mission drift

DL6/DL Ω systems can remain active while gradually pursuing the wrong mission. Persistent mission state, change logs, and external success criteria are therefore necessary. Autonomy does not make objective drift acceptable.

17. Research Program

A useful empirical program can proceed in six stages.

Stage 1 - Instrument intervention. Build logs that distinguish governance actions from cognitive assistance and record intervention type, duration, timing, and criticality.

Stage 2 - Calibrate task difficulty. Create difficulty vectors and T0-T6 task-family anchors with independent raters and measurable structural features.

Stage 3 - Measure frontiers. For each system, estimate $F_A(h,p)$ under standardized H0-H3 policies rather than reporting one autonomy score.

Stage 4 - Test longitudinal stability. Repeat tasks after restarts, environment changes, tool failures, and delayed dependencies to distinguish prompt-local success from persistent delegation.

Stage 5 - Evaluate organizations. Construct tasks where coordination is essential and measure whether O-DL exceeds the frontier of the strongest member while preserving verification quality.

Stage 6 - Test mission and charter levels. Evaluate long-horizon mission decomposition and, only after that is reliable, bounded-charter mission generation. DL Ω evidence should require repeated useful mission generation and external validation over time.

18. Discussion

Delegation Intelligence changes the question from "how autonomous is this agent?" to a more operational one: what level of cognitive responsibility can be transferred? The shift matters because autonomy is conditional on task difficulty. A system can be autonomous on easy tasks and highly dependent on human cognition on hard ones. The DL framework makes this gradient explicit.

The framework also emphasizes that less human intervention is not intrinsically better. Human governance can be desirable even for cognitively autonomous systems. Likewise, an agent that escalates correctly when evidence is missing can be more trustworthy than one that insists on H0 behavior by guessing. DL is therefore not a policy recommendation to remove people. It is a measurement of how much human cognition is required for successful completion under a chosen policy.

A second consequence is that delegation is not the same as developmental intelligence. A fixed but powerful agent may achieve high DL without learning. A self-improving agent may remain low DL if it cannot reliably manage complex work. This independence is useful: it allows systems to be compared without assuming that every desirable property lies on one ladder.

Finally, organizational delegation deserves independent measurement. As AI systems become multi-agent, the practical unit delegated to may be a persistent organization rather than one model. The organization must then be evaluated for decomposition, role selection, shared state, verification, coordination failures, and human management burden. O-DL is not the maximum DL of its members; it is the delegation capacity of the structure as a whole.

19. Conclusion

Delegation Intelligence is the capacity to convert human goals into independently verified outcomes while minimizing the human cognitive work required inside the task loop, conditional on task difficulty and reliability.

The framework is summarized by three objects:

Task difficulty: T0 atomic -> T6 open mission.

Human cognitive intervention: H5 continuous -> H0 none.

Delegation level: DL0 instruction-following -> DL6 mission autonomy -> DL Ω bounded-charter open-ended delegation.

Its central measurement is not the label itself but the delegation frontier:

$F_A(h,p)$ = the highest task difficulty system A completes with success probability $\geq p$ under intervention budget h .

This makes delegation falsifiable. A stronger intelligence must move the frontier toward harder tasks, lower cognitive intervention, or greater reliability - ideally all three - without hiding human work, redefining task difficulty, or self-certifying success.

The practical question can then be stated precisely:

How difficult a goal can I hand to this intelligence, how little must I think for it, and how often does the result survive independent verification?

That question is narrower than "How intelligent is it?" and, for many real deployments, more useful.

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